

Masrur Sarar

ID:- 23201003

Section:- 10

Assignment-2 CSE 422

1)

a) From the table we get,

$$P(S) = 0.25 + 0.15 + 0.02 + 0.03 = 0.45$$

$$P(C \cap S) = 0.25 + 0.02 = 0.27$$

$$P(P \cap S) = 0.25 + 0.15 = 0.4$$

If conditionally independent then,

$$P(C \cap P | S) = P(C | S) \times P(P | S)$$

Now,

$$P(C \cap P | S) = \frac{P(C \cap P \cap S)}{P(S)} = \frac{0.25}{0.45} = 0.56$$

$$\begin{aligned} P(C | S) \times P(P | S) &= \frac{P(C \cap S)}{P(S)} \times \frac{P(P \cap S)}{P(S)} \\ &= \frac{0.27}{0.45} \times \frac{0.4}{0.45} = 0.533 \end{aligned}$$

$$\therefore P(C \cap P | S) \neq P(C | S) \times P(P | S)$$

So cheat and pass are not conditionally independent

$$b) P(P) = 0.25 + 0.15 + 0.1 + 0.13 = 0.63$$

$$P(C) = 0.25 + 0.02 + 0.1 + 0.22 = 0.59$$

$$P(P \cap C) =$$

$$P(P \cap C) = 0.25 + 0.1 = 0.35$$

$$P(P \cup C) = 0.63 + 0.59 - 0.35 = 0.87$$

Here,

S = Study

P = Pass

C = cheat

2) C = Cold
CI = Cloudy
R = Rain

$$a) P(C) = 0.32 + 0.06 + 0.26 + 0.03 = 0.67 \quad P(C') = 0.12 + 0.04 + 0.1 + 0.07 = 0.33$$

$$b) P(CI' | R' \cap C') = \frac{P(CI' \cap R' \cap C')}{P(R' \cap C')} = \frac{0.7}{0.7 + 0.4} = 0.64$$

$$c) P(R' | CI') = \frac{P(R' \cap CI')}{P(CI')} = \frac{0.03 + 0.07}{0.26 + 0.03 + 0.1 + 0.07} = 0.22$$

$$d) ~~P(R' \cap C)~~ P(R') = 0.06 + 0.04 + 0.03 + 0.07 = 0.2$$

$$P(CI) = 0.32 + 0.06 + 0.12 + 0.04 = 0.54$$

$$P(R' \cap CI) = 0.06 + 0.04 = 0.1$$

$$P(R' \cup CI) = 0.2 + 0.54 - 0.1 = 0.64$$

$$3) a) P(F | L) = \frac{P(F \cap L)}{P(L)} = \frac{0.15}{0.24 + 0.15 + 0.15} = 0.28$$

$$b) P(R | C) = \frac{P(R \cap C)}{P(C)} = \frac{0.1}{0.24 + 0.1} = 0.29$$

$$c) P(F \cap C) = 0$$

$$d) P(R \cup L) = 0.24 + 0.15 + 0.15 + 0.1 + 0.1 + 0.26 = 1$$

$$e) P(F | R) = \frac{P(F \cap R)}{P(R)} = \frac{0.1}{0.1 + 0.1 + 0.26} = 0.21$$

$$P(F) = 0.1 + 0.25 = 0.25$$

$$\therefore P(F | R) \neq P(F)$$

So playing Football does ~~not~~ dependent on being right handed.

Here,

F = Football

C = Cricket

O = Other

L = Left-handed

R = Right-handed

4)

$$P(!b) = \frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8} = \frac{4}{8} = 0.5$$

$$P(!a \cap !b) = \frac{1}{8} + \frac{1}{8} = \frac{2}{8} = 0.25$$

$$P(!a|!b) = \frac{P(!a \cap !b)}{P(!b)} = \frac{0.25}{0.5} = 0.5$$

1) a)

~~$$P(L|D) = \frac{P(L \cap D)}{P(D)} = \frac{0.96 \times 0.02}{0.0192} = \frac{0.0192}{0.0192}$$~~

~~$$P(L'|D) = \frac{P(L' \cap D)}{P(D)} = 0$$~~

Here,

D = Detect lie

L = Liar

$$P(L|D) = \frac{P(D|L)P(L)}{P(D)} = \frac{0.96 \times 0.02}{0.0192} = \frac{0.0192}{0.0192}$$

$$P(L'|D) = \frac{P(D|L')P(L')}{P(D)} = \frac{0.05 \times 0.98}{0.0192} = \frac{0.049}{0.0192}$$

$$\frac{P(L'|D)}{P(L|D)} = \frac{0.049}{0.0192} = 2.55$$

$$\therefore P(L'|D) > P(L|D)$$

The person is more likely to be not liar

$$b) P(D) = P(D|L)P(L) + P(D|L')P(L')$$

$$= 0.96 \times 0.02 + 0.05 \times 0.98 = 0.0682$$

$$P(L'|D) = \frac{P(D|L')P(L')}{P(D)} = \frac{0.05 \times 0.98}{0.0682} = 0.7185$$

$$2) \\ a) P(\text{Yes} | S, M, N, T) = \frac{P(S | \text{Yes}) \times P(M | \text{Yes}) \times P(N | \text{Yes}) \times P(T | \text{Yes}) \times P(\text{Yes})}{P(S, M, N, T)} \\ = \frac{\frac{2}{5} \times \frac{3}{5} \times \frac{4}{5} \times \frac{3}{5} \times \frac{5}{8}}{P(S, M, N, T)} = \frac{0.072}{P(S, M, N, T)}$$

$$P(\text{No} | S, M, N, T) = \frac{P(S | \text{No}) \times P(M | \text{No}) \times P(N | \text{No}) \times P(T | \text{No}) \times P(\text{No})}{P(S, M, N, T)} \\ = \frac{\frac{3}{3} \times \frac{2}{3} \times \frac{6}{3} \times \frac{1}{3} \times \frac{3}{8}}{P(S, M, N, T)} = 0$$

Here,
S = Sunny
M = Mid
N = Normal
T = True

$$\therefore P(\text{Yes} | S, M, N, T) > P(\text{No} | S, M, N, T)$$

So the player is more likely to play Tennis

$$b) P(Y | O, H) = \frac{P(O | Y) \times P(H | Y) \times P(Y)}{P(O, H)} = \frac{\frac{2}{5} \times \frac{0}{5} \times \frac{5}{8}}{P(O, H)} = 0 \times \frac{1}{20} = 0$$

Here,
Y = Yes
N = No
O = Overcast
H = Hot

$$P(N | O, H) = \frac{P(O | N) \times P(H | N) \times P(N)}{P(O, H)} = \frac{\frac{0}{3} \times \frac{1}{3} \times \frac{3}{8}}{P(O, H)} = 0 \times \frac{1}{24} = 0$$

Both become zero but $P(Y | O, H)$ is higher if we don't multiply zero at the end.

But the final result being the same we can say that the probability of playing and not playing Tennis are the same.

3) X	Y	$\hat{Y}$	$(Y - \hat{Y})$
2	14	7	7
4	29	13	16
6	42	19	23

$$a) \text{Error} = \sum_{i=1}^n \frac{(Y - \hat{Y})^2}{n}$$

$$= \frac{1}{3} (7^2 + 16^2 + 23^2)$$

$$= 278$$

$$b) \frac{\partial L}{\partial m} = -\frac{2}{n} \sum_{i=1}^n (Y - \hat{Y})X = \frac{2}{3} \times \{ (7 \times 2) + (16 \times 4) + (23 \times 6) \} = -144$$

$$\frac{\partial L}{\partial b} = -\frac{2}{n} \sum_{i=1}^n (Y - \hat{Y}) = \frac{2}{3} \times \{ 7 + 16 + 23 \} = -30.67$$

$$m_{\text{new}} = 3 - 0.01 \times (-144) = 4.44$$

$$b_{\text{new}} = 1 - 0.01 \times (-30.67) = 1.3$$

4)

$$a) P(E) = \frac{9}{16}, P(E') = \frac{7}{16}$$

$$H(E) = -\frac{9}{16} \log_2 \left(\frac{9}{16} \right) - \frac{7}{16} \log_2 \left(\frac{7}{16} \right) = 0.988$$

$$b) H(E|Y) = 0.96$$

$$H(E|S) = 0.81$$

$$H(E|R) = 1$$

$$H(E|G) = 0.92$$

$$H(E|L) = 0.95$$

$$H(E|I) = 0.81$$

$$H(E|C) = 0.95$$

$$H(E|Si) = 0.88$$

$$H(E|Sh) = 0.953$$

Here,

E = Edible

Y = Yellow

G = Green

S = Small

L = Large

R = Regular

I = Irregular

C = colour

Si = size

Sh = Shape

$$\text{Information gain (colour)} = 0.988 - 0.95 = 0.038$$

$$\text{Information gain (Size)} = 0.988 - 0.88 = 0.108$$

$$\text{Information gain (shape)} = 0.988 - 0.953 = 0.035$$

$\therefore$ Size is better feature than other.

5)

$$P(\text{Pass}) = 0.352 + 0.152 + 0.101 + 0.052 = 0.657$$

$$P(\text{Fail}) = 1 - 0.657 = 0.343$$

$$H(R) = 0.928$$

$$P(P_r) = 0.554$$

$$P(P_r') = 0.446$$

$$P(\text{Pass} | P_r) = 0.82$$

$$P(\text{Fail} | P_r) = 0.18$$

$$P(\text{Pass} | P_r') = 0.46$$

$$P(\text{Fail} | P_r') = 0.54$$

$$H(R | P_r) = 0.68$$

$$H(R | P_r') = 0.99$$

$$H(R | P) = 0.83$$

$$IG(P) = 0.098$$

$$P(G) = 0.626$$

$$P(B) = 0.374$$

$$P(\text{Pass} | G) = 0.805$$

$$P(\text{Fail} | G) = 0.195$$

$$P(\text{Pass} | B) = 0.41$$

$$P(\text{Fail} | B) = 0.59$$

$$H(R | G) = 0.71$$

$$H(R | B) = 0.97$$

$$H(R | E) = 0.81$$

$$IG(E) = 0.118$$

Here,

R = Result

P = Preparation

E = Exam

P_r = Prepared

P_r' = Not Prepared

G = Good

B = Bad

$$\therefore IG(E) > IG(P)$$

So Exam is the better feature

6)

$$a) Z = w_1 x_1 + w_2 x_2 + b = 0.5 \times 1 + -0.25 \times 2 + 0 = 0$$

$$y' = \frac{1}{1 + e^{-z}} = \frac{1}{1 + e^0} = 0.5$$

b)

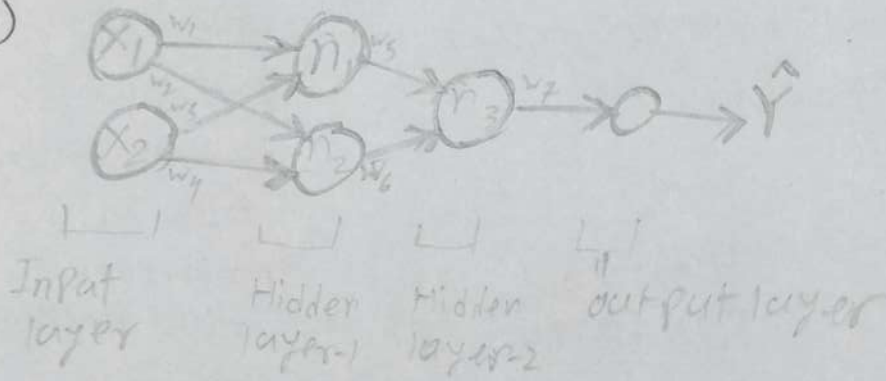
$$\frac{\partial L}{\partial w_1} = (y' - y) x_1 = (0.5 - 0) \times 1 = 0.5$$

$$w_{1 \text{ new}} = w_1 - \alpha \frac{\partial L}{\partial w_1} = 0.5 - 0.1 \times 0.5 = 0.45$$

c) Decaying learning rate allows to take large steps at the start and to approach minimum quickly and short steps later to settle precisely into the minimum.

7)

a)



b)

$$a'_1 = 0.5 \quad w_5 = 0.1$$

$$a'_2 = 0.5 \quad w_6 = 0.5$$

$$z = a'_1 w_5 + a'_2 w_6 + b = 0.5 \times 0.1 + 0.5 \times 0.5 + 0 = 0.3$$

$$a_1^2 = \sigma(z) = \frac{1}{1 + e^{-z}} = \frac{1}{1 + e^{-0.3}} = 0.574$$

c) Hidden Layer 1 has 2 output because it learns separate low-level features from the inputs. Hidden layer 2 has only 1 output because it aggregates those features into a single higher-level representation to make a prediction. The more the ~~net work~~ ^{neuron} the more the ~~out put~~ ^{activation} output.